

# PROJECT: ROBOT NAVIGATION USING FINITE AUTOMATA

## Objective

The goal of this project is to:

- design a **formal command language** for a robot navigation system
- model the language using a **finite automaton (FA)**
- implement a **visual simulation** that executes valid commands

## Description

You will design a robot that moves on an **8×8 grid**, starting at the bottom left position (0,0) and facing North. The robot follows a sequence of commands to move (forward or backward), turn (left or right), pick up an object, drop it, and recharge energy. The robot has a **limited energy capacity**. Each movement command consumes 1 unit of energy. The robot has a maximum of 3 energy units. If the robot attempts to move without energy, the command sequence is considered invalid.

Your command language must enforce rules such as:

1. The sequence must start with START and end with STOP
2. The robot must perform at least one movement (forward or backward)
3. Picking up object must occur before dropping the object, and the robot cannot pick twice without dropping
4. The robot must complete at least one pick-drop task before STOP
5. No more than 2 consecutive turns (left or right)
6. Movement is only allowed when energy is available, and recharge resets energy to full

These rules must be enforced using your **finite automaton design**, not only in the simulation.

## Tasks

- **Language & FA Design**
  - define the alphabet and rules
  - construct a DFA or NFA that accepts only valid sequences
- **Robot Simulation (Visual)**
  - implement a program that:
    - validates input using your automaton
    - visually simulates the robot on the grid step by step

## Deliverables

- report (language, FA, explanation)
- source code (simulation)
- presentation

**Additional Rules:**

7. No immediate reverse movement. The robot cannot move forward immediately after moving backward, or backward immediately after forward.
8. The robot must complete at least two pick–drop tasks (*overwriting Rule #4*).
9. The robot can only recharge when its energy is exactly 0.
10. The robot cannot turn left or right twice in a row (*overwriting Rule #5*).
11. Turning also consumes energy. The max energy capacity is now 5.
12. The robot must perform at least one counter-clockwise loop: (forward -> left) x4
13. The robot must not perform any clockwise loop: (forward -> right) x4
14. At any point in the command sequence, the absolute difference between the number of left turns and right turns must not exceed 2.